

Revision Notes for Class 7 Maths Chapter 6 Number Play

In this chapter, "Number Play", students explore how numbers can represent different properties and patterns in our daily lives. The chapter begins with an interesting activity involving children standing in a sequence and calling out numbers. Each child calls out the number of children in front of them who are taller than them. This simple rule connects the idea of ordering and arrangement without knowing the actual heights. By experimenting with different arrangements, students understand how numbers can describe positions and relationships.

Picking Parity: Even and Odd Numbers

Parity is a key topic covered in this chapter. Parity refers to whether a number is even or odd. Students learn that even numbers can be arranged in pairs without anything left, while odd numbers always have one unpaired item. Through activities, examples, and puzzles, the chapter shows that:

- The sum of two even numbers is always even.
- The sum of two odd numbers is also always even.
- The sum of an even and an odd number is always odd.
- Adding any odd number of odd numbers will always result in an odd number.
- Adding any even number of odd numbers will result in an even number.

For example, when students try to fill 5 boxes with odd number cards to reach a sum of 30 (an even number), they quickly learn this is impossible because five odd numbers add up to an odd sum. The idea extends to practical situations, such as figuring out if two siblings, whose ages differ by one, can have their ages sum to an even number (it is impossible, because consecutive numbers are always one even and one odd, and their sum is odd).

Formulas and Algebraic Expressions for Parity

The chapter introduces students to simple algebraic formulas to generate even and odd numbers. For instance, the n th even number is given by $2n$, and the n th odd number by $2n - 1$. Students also examine expressions like $3n + 4$, noticing that depending on n , such expressions can yield even or odd results. They are encouraged to invent more such expressions and find ones that generate only even numbers (like $100p$, $48w - 2$) or only odd numbers (like $2m + 1$, $4k - 1$).

Practical exercises reinforce these concepts by asking students to find the parity of numbers in a grid, the sum or difference of different types of numbers, and the parity patterns in real-life number arrangements.

Working with Grids and Magic Squares

Students engage with grids, where the task is to fill a 3×3 square using numbers 1 to 9 without repetition, focusing on the sums of rows and columns. They learn:

- The minimum possible sum for any row (or column) is $1 + 2 + 3 = 6$; the maximum is $7 + 8 + 9 = 24$.
- If any row or column has a sum outside this range, it means the grid cannot be filled with distinct numbers from 1 to 9 to satisfy those sums.
- When all rows and columns have the same sum, and every number from 1 to 9 is used, the grid is called a magic square.
- For the 3×3 magic square with numbers 1–9, the "magic sum" (sum of each row, column, or diagonal) is always 15.
- The center of the standard 3×3 magic square is always 5, and the corners cannot have 1 or 9.

Students also see that magic squares have a long history, being recorded in many cultures, such as ancient China (the Lo Shu Square) and India (e.g., the Chautīsā Yantra

of Khajuraho, where the magic sum is 34 for a 4×4 square). Magic squares with other sets of consecutive numbers can be constructed using simple transformations.

Patterns, Sequences, and the Virahāṅka–Fibonacci Numbers

A major section is devoted to the famous Virahāṅka–Fibonacci sequence: 1, 2, 3, 5, 8, 13, 21, 34, ... Students learn how this sequence arises from poetry in ancient India, where scholars counted the number of rhythms possible with given beats (as sums of 1's and 2's). The rule is simple: Each term is the sum of the two preceding numbers. For example, to find the number of ways to fill 5 beats, add the ways for 4 and 3 beats ($5 = 3 + 2$). This idea can be extended, and students try to write the next numbers in the sequence.

This sequence is seen in art, music, poetry, nature (like the number of petals in flowers), and mathematics worldwide. Fun pattern recognition questions are added, like checking if each number is odd or even, or how steps increase in the sequence.

Cryptarithms: Digits in Disguise

The chapter introduces “cryptarithms,” mathematical puzzles where digits are replaced by letters. Each letter stands for a unique digit (0–9), and students must deduce the correct substitution. For example, if $T + T + T = UT$ and $T = 5$, then $UT = 15$. Many such puzzles are shared, challenging students’ logic and arithmetic skills. This section sharpens reasoning and problem solving by working backwards from information presented in coded form.

Practice, Reasoning, and Applications

Throughout the chapter, “Figure it Out” activities ask students to apply concepts to new situations: finding the parity of algebraic expressions, exploring possibilities in grids, or reasoning about number statements (such as the effect of toggling a switch an odd number of times, or understanding the sum of page numbers in a set of sheets).

The chapter concludes by summarizing its key ideas. Students have seen that parity helps in solving puzzles, magic squares reveal beautiful mathematical patterns, and the

Virahāṅka–Fibonacci sequence connects math to art and nature. By practicing these concepts, students strengthen their ability to spot patterns, reason logically, and approach problems creatively.

Key Points from the Chapter

- Understanding how sequences and arrangements can be described using numbers alone.
- Mastering the properties of even and odd numbers and their sums/products.
- Learning how to form algebraic expressions for sequences of numbers.
- Exploring grids and magic squares, including methods to check if certain arrangements are possible.
- Discovering the Virahāṅka–Fibonacci sequence as it appears in poetry, music, art, and science.
- Strengthening logic and deduction skills by solving cryptarithms and reasoning problems.